

Data Cleaning & Preprocessing using Python

Objective:

To clean and preprocess raw dataset and transform it into an analysis-ready format by handling missing values, duplicates, and inconsistencies.

Steps Performed:

- 1 Loaded dataset using pandas
- 2 Performed initial data inspection (info, head)
- 3 Removed duplicate records
- 4 Dropped irrelevant columns
- 5 Handled missing values using median/mode
- 6 Converted data types (string to numeric/datetime)
- 7 Standardized text formats (lowercase, trimming spaces)
- 8 Validated cleaned dataset

Tools Used:

- 1 Python
- 2 Pandas
- 3 NumPy
- 4 VS Code / Anaconda

Output:

The cleaned dataset is saved as 'cleaned_data.csv' and is ready for further analysis.

Conclusion:

The dataset was successfully cleaned and preprocessed by removing inconsistencies, handling missing values, and standardizing formats. This ensures accurate and reliable analysis in future tasks.